

# **What's So Important About Infrastructure?**

Unicenter Conference  
Session Code: UIT15P

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# Agenda

- Change versus continuity: who cares?
- Fragmented IT infrastructure issues
- Choosing an IT infrastructure
- Options, implications, and conclusions

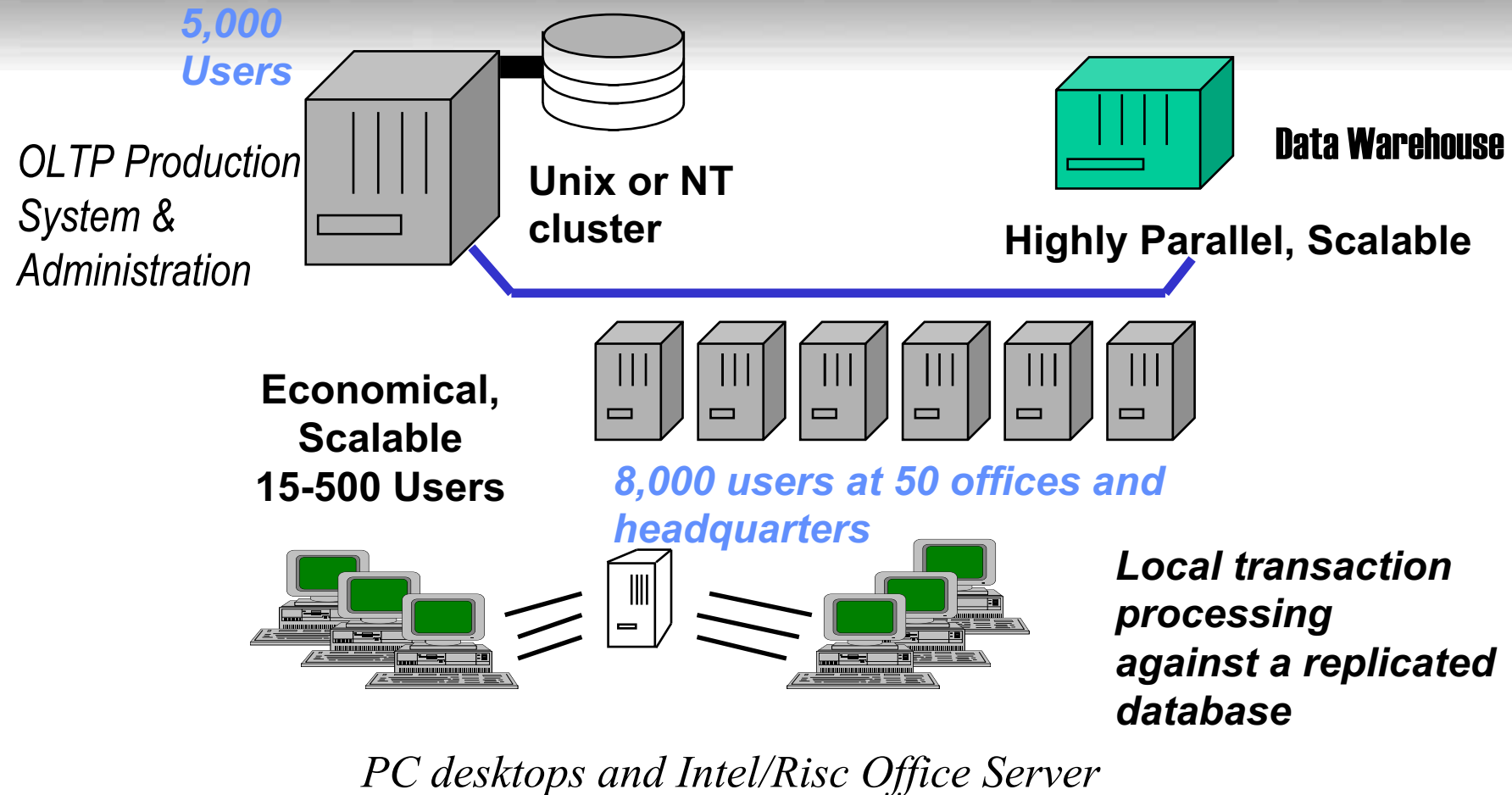
# IT Spending: New Acquisition versus Ongoing Operations

- U.S. enterprises spend ~3% of revenue on IS
- ~85% of IS spending is for ongoing operations — ~15% for new application acquisitions
- CEO looks at 15% for new applications as incremental change & business driver
- CFO and VP IS/CIO look at 85% for ongoing operations as area to be closely managed and controlled

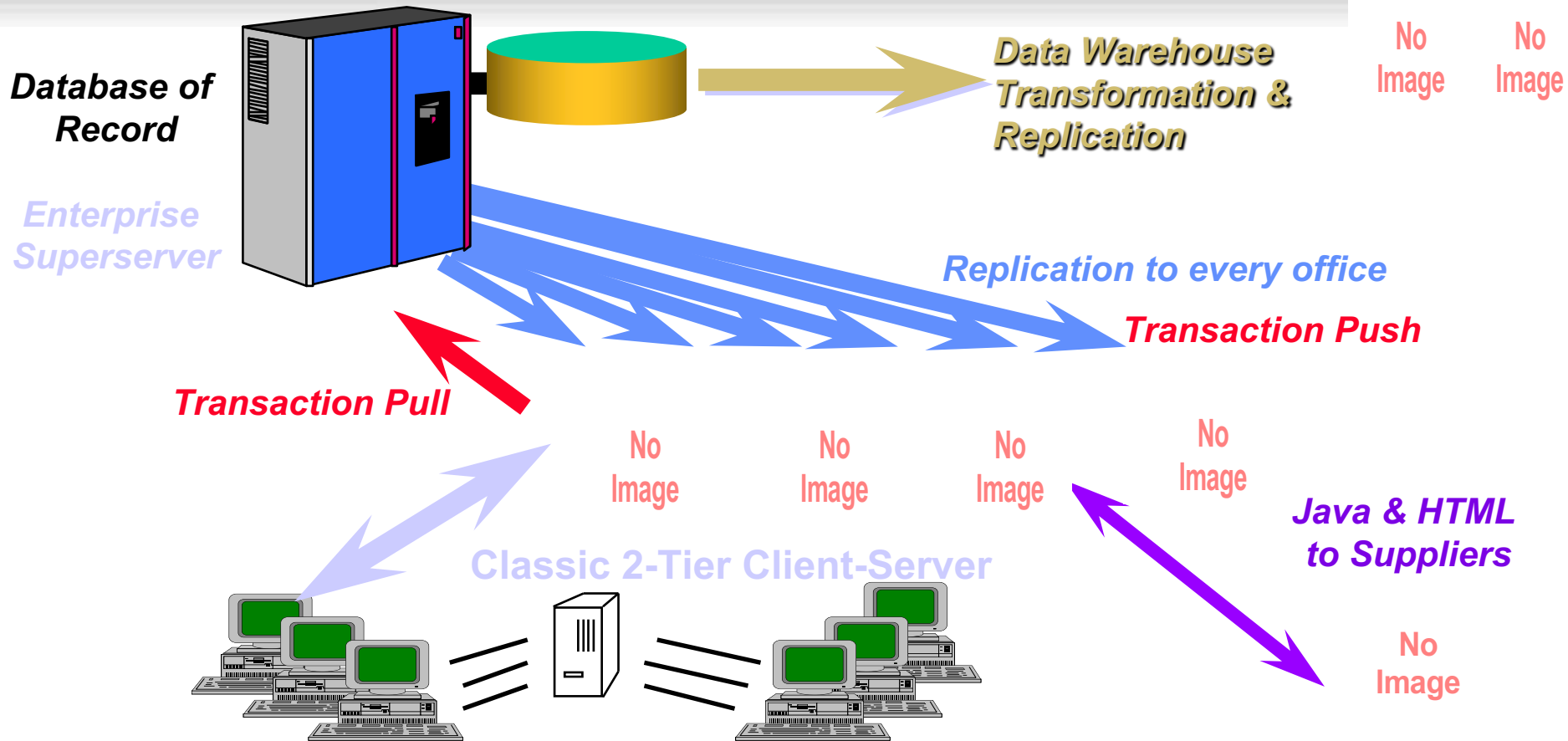
# IT-Driven Business Imperatives

- Increase customer satisfaction
- Empower fewer users
- Increase line access to data
- Increase operational flexibility
- Shorten application development cycle time

# Business Necessity Meets IS Reality: Insurance Industry



# IS Reality: *Enormous Complexity*

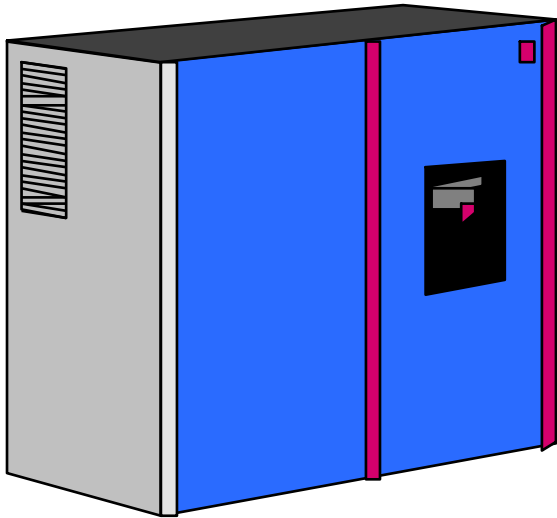


# IT Infrastructure Fragments

- Enterprise mainframe
- Geographical islands of automation
- LANs become Intranets
- Data warehousing
- Client/server meets the Web
- E-business is the tip of an iceberg

*But the IT Dept. Has to Sweat the Technology Details ...*

# What Happened to the Mainframe Data Center?



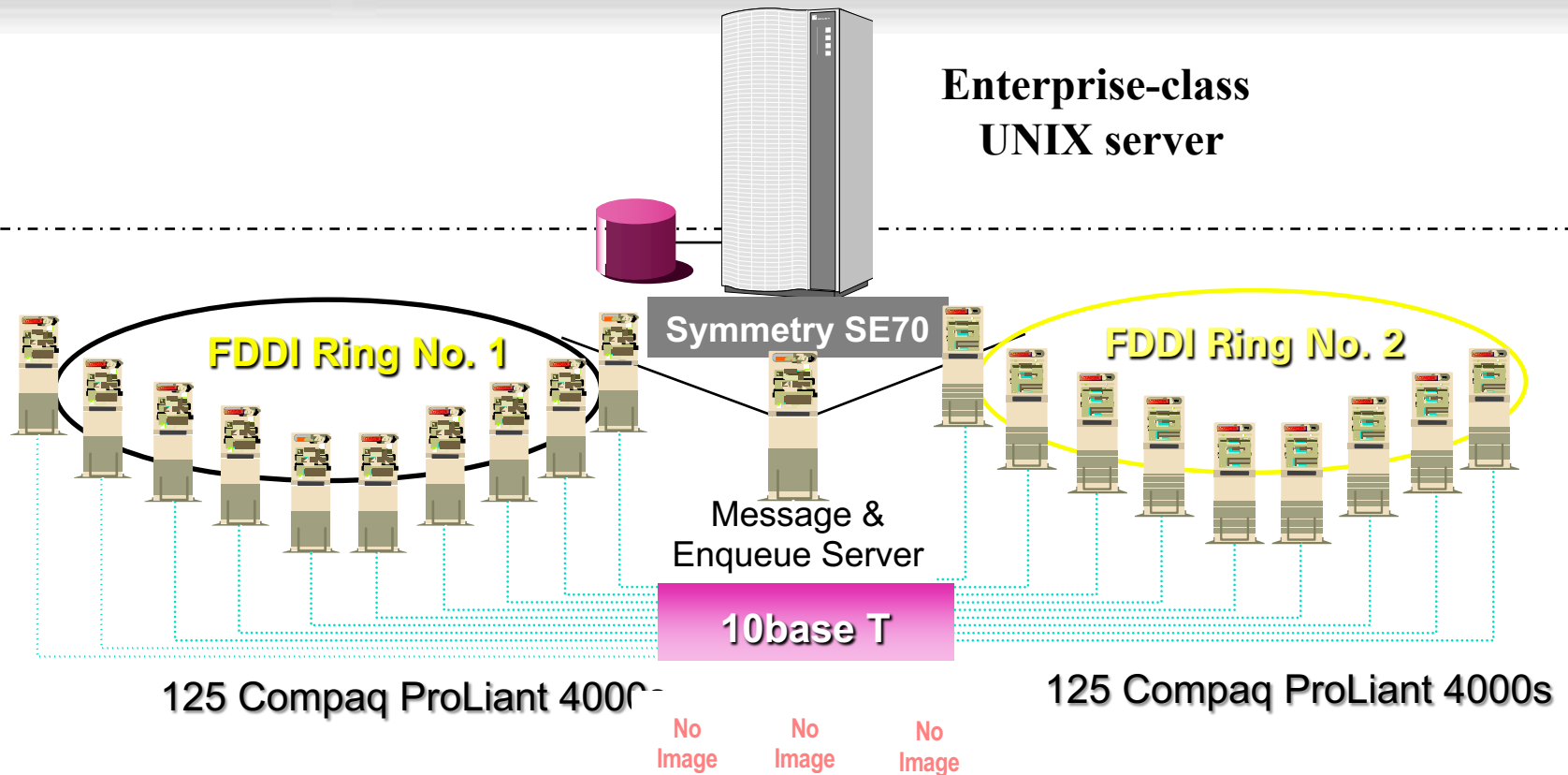
- Thoroughly understood
- Predictable reliability
- Outstanding operational efficiency, productivity, and manageability

But ...

- Hasn't seen a new ISV application in 7 years
- Is being surrounded or supplanted

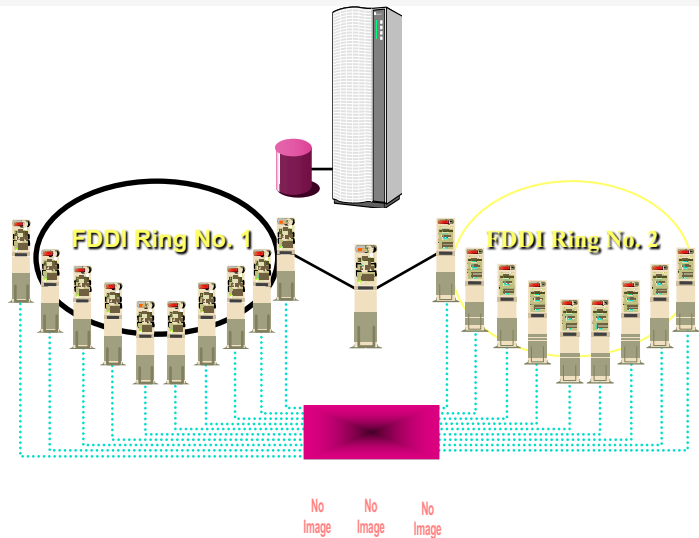


# New-Breed Enterprise Mainframe



LANs

# New-Breed Enterprise Mainframe

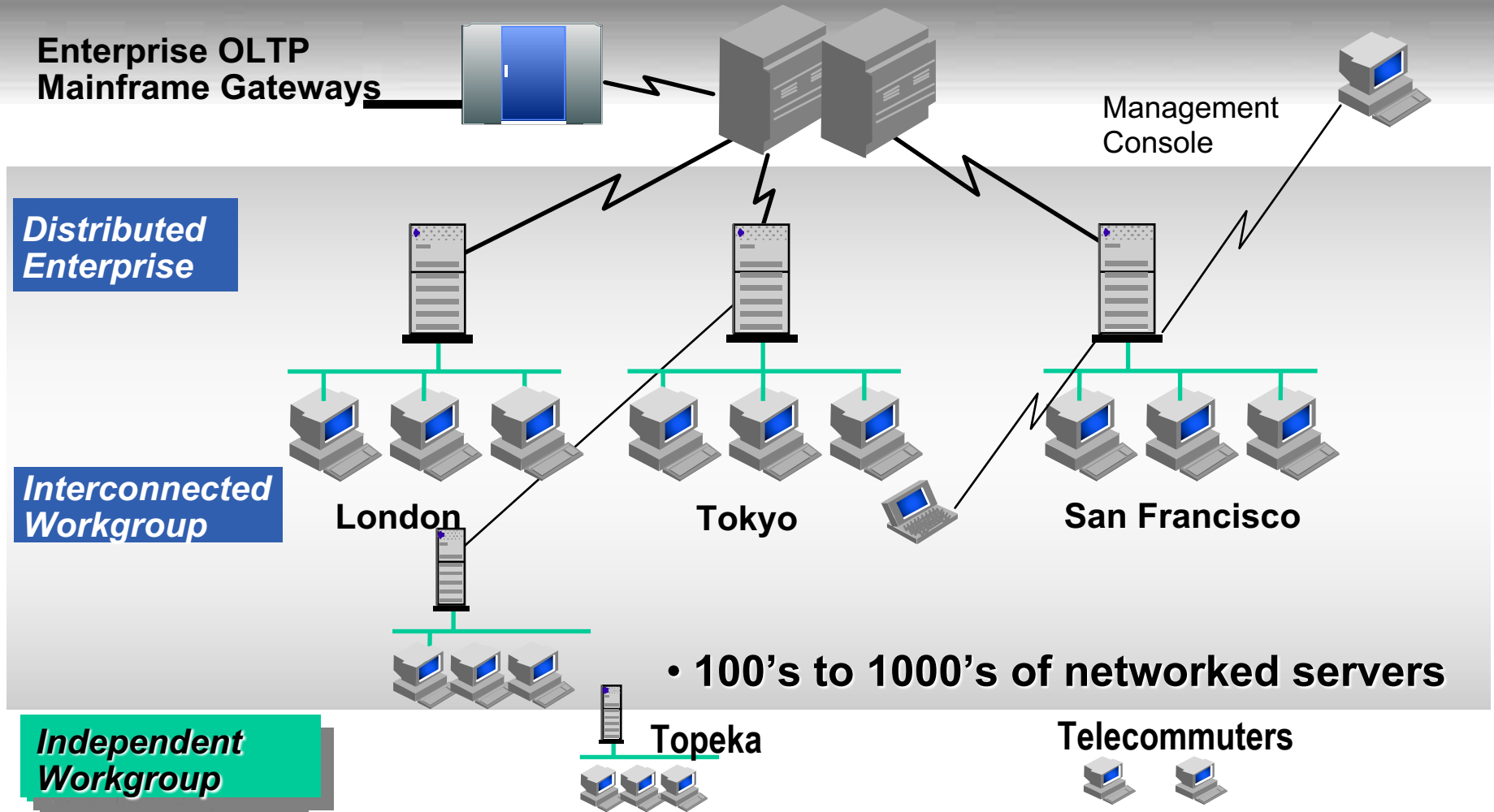


- N-tier architecture
- Reliability, operability, manageability constraints
- Multi-vendor control issues

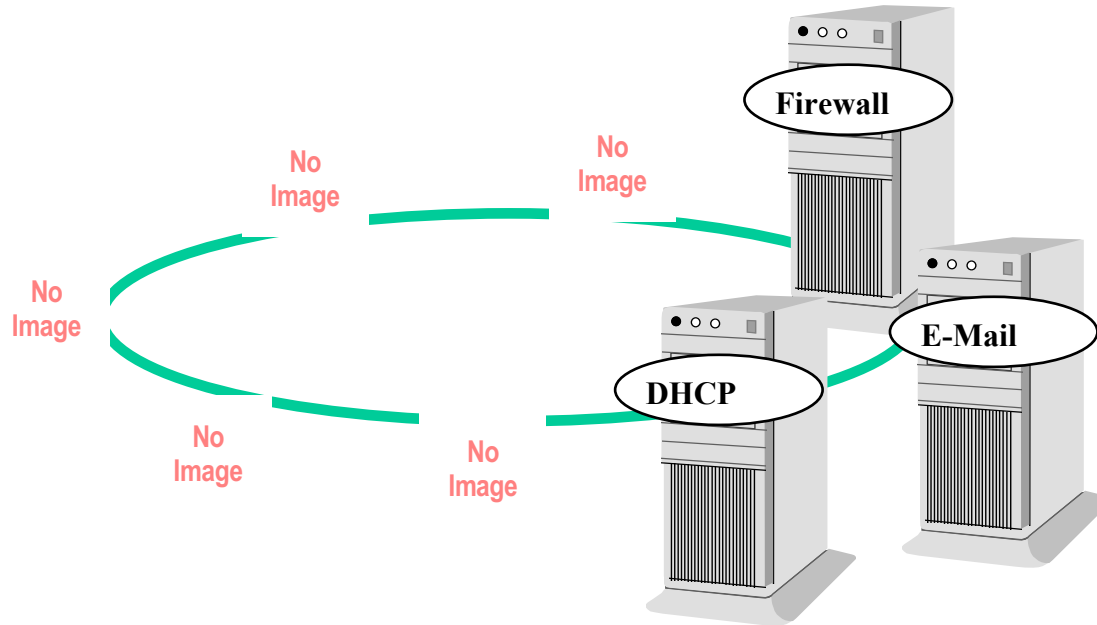
But ...

- Running today's mission-critical enterprise applications: a paradox

# Geographically Distributed Enterprise



# Shift from NOS to Intranet



## New LAN Functions:

- Applications
- Decision support
- Mainframe connectivity
- Exchange server
- Internet proxy
- Firewall
- Systems management
- Development
- WINS & DHCP

*“95% of Fortune 1000 Will Have Intranets By 2000” --  
Forrester*

# Data Warehousing: “Great Idea, Hard to Implement”

*OLTP Transactions*

*Transformation*

*Enterprise DW Server*

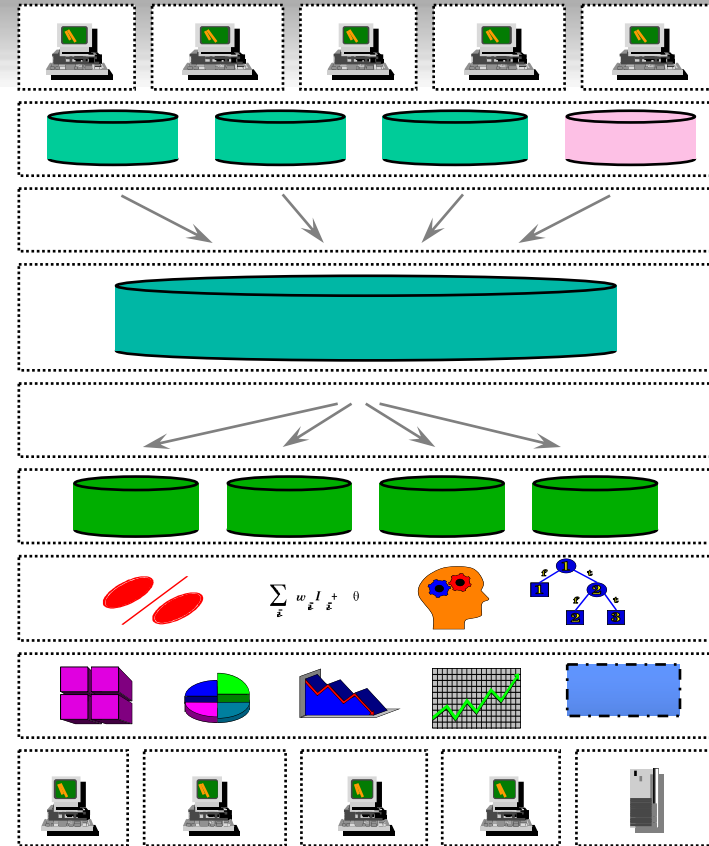
*Transformation*

*Data Mart Servers*

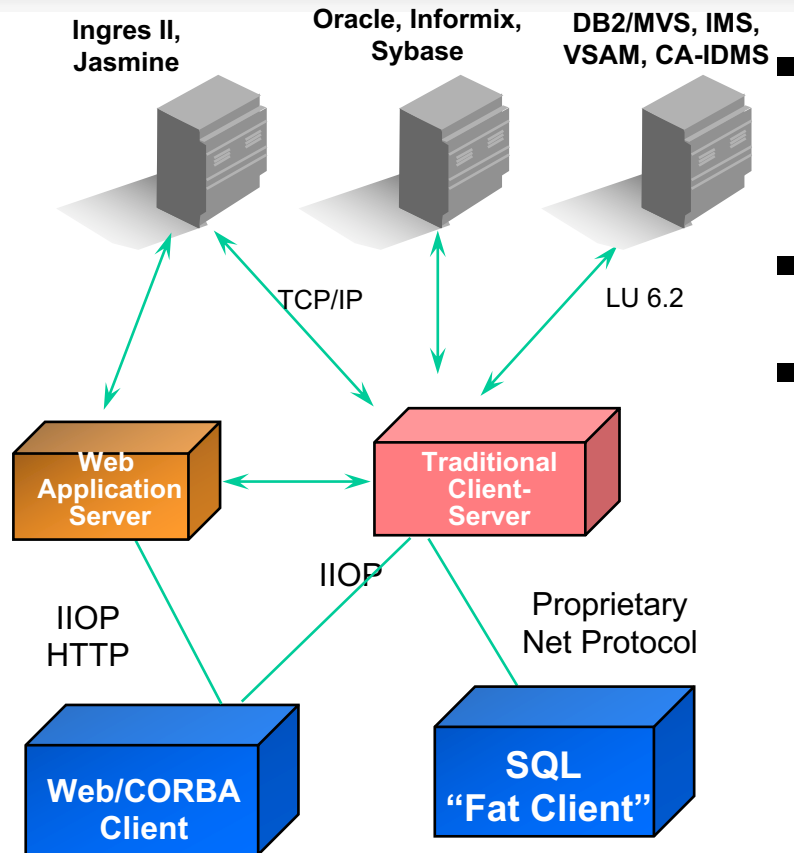
*Data Mining Tools*

*Data Visualization Tools*

*Decision Support Clients*



# Client/Server Integration with Web Applications



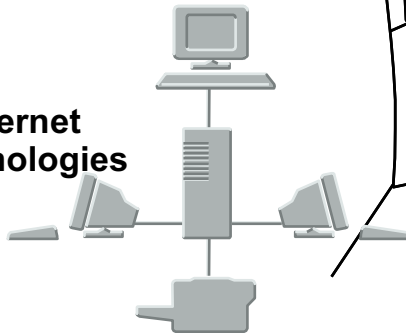
- Just as IS masters client/server technology ...
- Along comes the Web!
- Issues:
  - Legacy C/S integration
  - Evolving but “not there” standards
  - N-tier complexities
  - Programmer unfamiliarity

# CEO's Infinite Resource Planning Vision

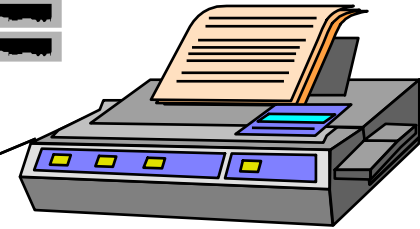


**Enterprise Systems**

**Internet Technologies**

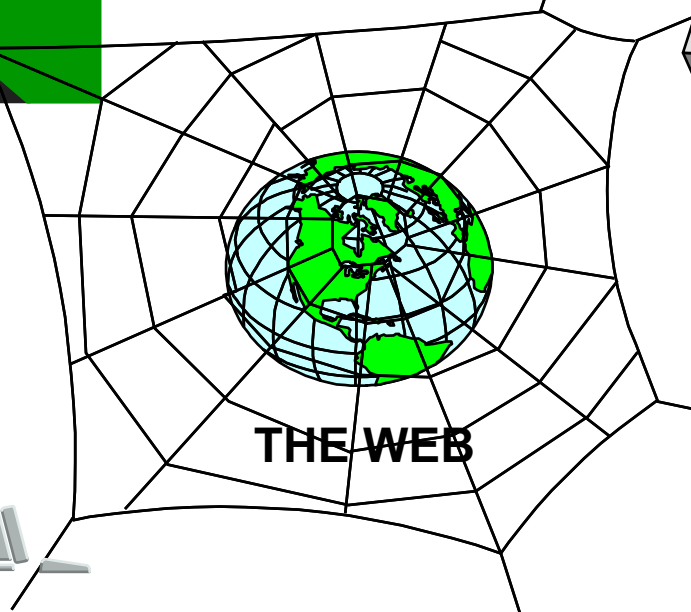


**Planning & Forecasting**

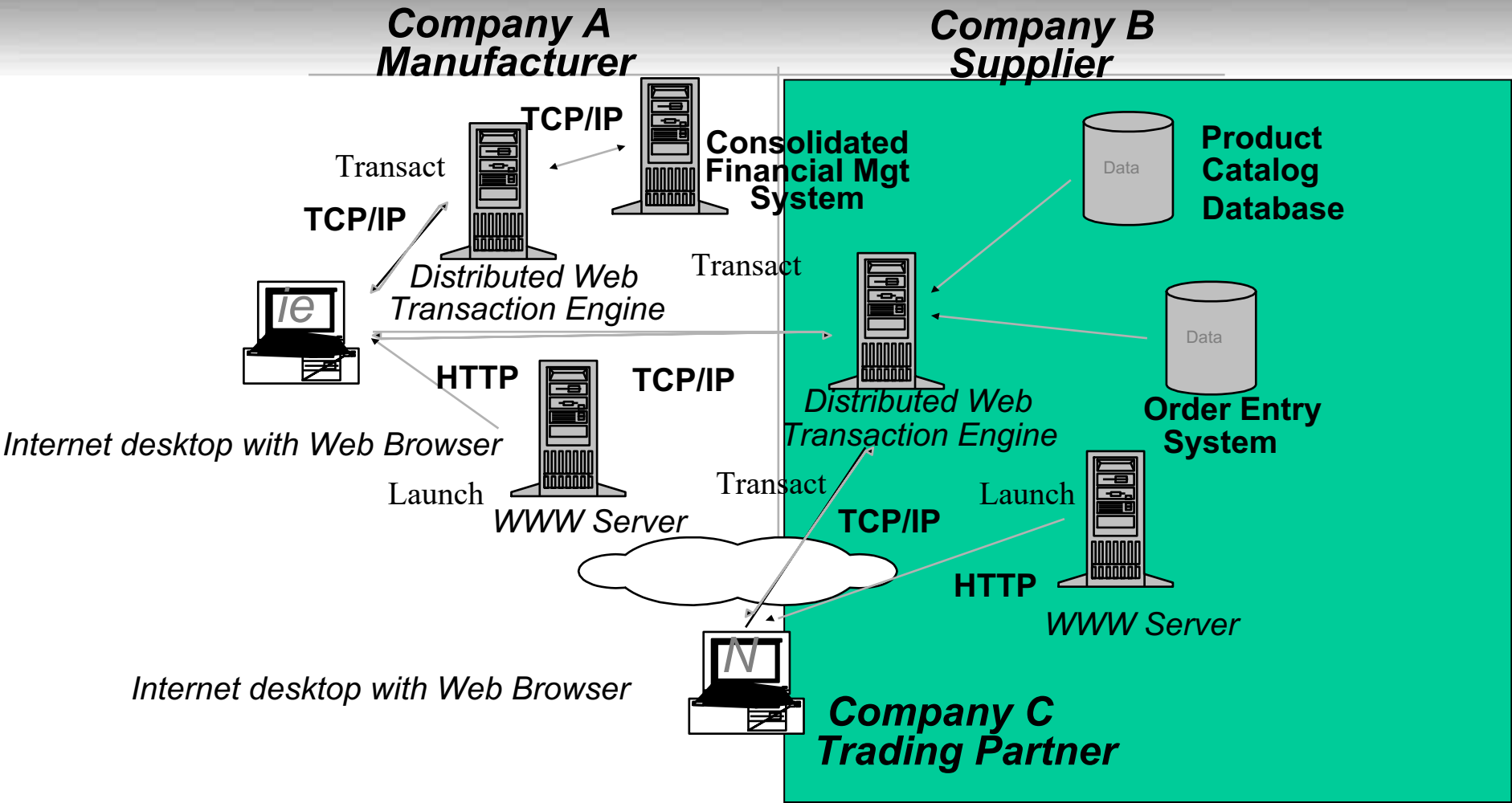


**EDI**

**Commodity Brokerage**



# Virtual Internet Computing: Reality is Complex, Multi-Company





# Building a Common Infrastructure: Management Options

- Do nothing
- Outsource
- Standards-based
- Architecture
- Vendor-based

# Do Nothing

- *Take no active steps to build a manageable common IM and EM infrastructure*
- Benefits
  - Just keep on doing what you're doing
- Implications
  - Fragmentation gets worse
  - Ability to disentangle and react diminishes
  - If you buy this, today's speakers have failed abjectly!

# Outsource

- *Outside service provider handles IT services*
- Benefits
  - Provider assumes future-shock risks
  - Establishes cost-for-service basis
  - One direction for finger pointing
- Implications
  - Will provider upgrade your technology base fast enough?
  - What key IS skills will atrophy?
  - Best suited for targeted cost-benefits (e.g., telecomm, “out-tasking”)

# Standards-based Infrastructure

- *Using industry standards, create an approved set of hardware, software, and networking technologies*
- Benefits
  - Interoperability yields easier systems integration
  - Less vendor technology lock-in
- Implications
  - Standards come slower; often are incomplete; always behind leading edge
  - Do not always provide the optimal solution
  - IS still has to build/integrate on standard platforms

# Architecture

- *Choose the key technologies for an architecture, then the best-of-breed products to implement it*
- Benefits
  - Architecture fits your enterprise's requirements (No “one size fits all” solution)
  - Best-of-breed products can change over time while preserving the investment in the architecture
- Implications
  - Requires intense discipline to avoid “exceptions”
  - What is best for the enterprise may not be best for the line-of-business system buyer

# Vendor-based

- *Choose a vendor's solutions and build on those*
- Benefits
  - Vendor endures the technology shocks
  - Focused on rolling out leading-edge solutions
  - Some supply value-added technology integration
- Implications
  - No *single* vendor can supply all the infrastructure
  - If the vendor fails, so do you
  - At the mercy of the vendor's technology curve
  - Nevertheless, an attractive choice

# Aberdeen's Rating of Management Options

- Do nothing      🐼 Bad idea
- Outsource      🐼 Parts, yes. The whole, never.  
Staff, never.
- Standards-based      🐼 Remember DCE, DME & OSI?
- Architecture      🐼 Good idea. Works best with  
strong central IS.
- Vendor-based      🐼 Good idea. Requires excellent  
relationship and basis for trust.

# Aberdeen's Conclusions

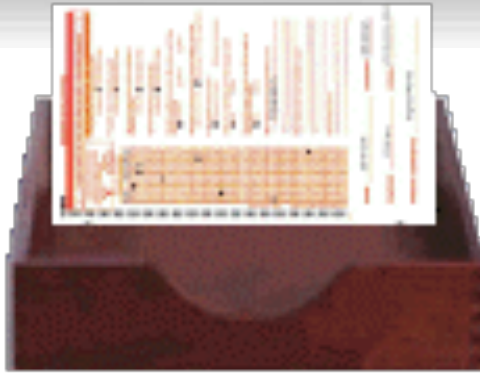
- IT-driven business is the wave of the present
- Information technology is exploding -- threatening to fragment IS development, operations, and networks into unmanageable pieces
- An architected, vendor-backed *common* infrastructure yields the greatest benefits
  - Must support both information and enterprise management needs



# Questions & Answers



# Session Evaluation Form



After completing  
your evaluation ...

... place it in the basket at the  
back of the room.

**SESSION EVALUATION**

PLEASE COMPLETE THIS FORM AND RETURN TO SESSION MONITOR

**MARKING DIRECTIONS**  
• Use blue or black ink only.  
• Fill circle completely.  
RIGHT      WRONG

**FILL IN THE CIRCLE OF THE APPROPRIATE RATING**  
Comprehension of subjects covered  
EXCELLENT (1) (2) (3) (4) (5) (6) (7) (8) (9) (10) (11) (12) (13) (14) (15) (16) (17) (18) (19) (20)

Practical value of material covered  
EXCELLENT (1) (2) (3) (4) (5) (6) (7) (8) (9) (10)

Speakers preparedness & knowledge of subjects covered  
EXCELLENT (1) (2) (3) (4) (5) (6) (7) (8) (9) (10)

Quality & Quantity of visual aids  
EXCELLENT (1) (2) (3) (4) (5) (6) (7) (8) (9) (10)

Speakers ability to control discussions and keep session moving  
☐ Excellent ☐ Good ☐ Poor

Allocation of time among various subjects covered  
☐ Too much time ☐ Appropriate ☐ Not enough time

Is this session part of your:  
☐ Primary conference ☐ Another conference

Were your expectations of this session met?  
☐ Yes ☐ No

Should this type of session be offered again at next year's conference?  
☐ Yes ☐ No

What topics were most valuable to you?  
1. \_\_\_\_\_  
2. \_\_\_\_\_

What topics were least valuable to you?  
1. \_\_\_\_\_  
2. \_\_\_\_\_

General Comments: \_\_\_\_\_

NAME \_\_\_\_\_ SESSION # \_\_\_\_\_

COMPANY \_\_\_\_\_ PHONE # \_\_\_\_\_

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